Lecture 4 Notes

Quantitative Data Analysis

Introduction

- Last time we inspected and described our data
- Today, we will expand on this and look at how to visualise the data
- Data visualisation is a crucial step in any analysis
- Minimally, it will help us to understand our data and spot patterns
- In some cases, a well put-together visualisation will be sufficient to make the point
- Further analysis may not be necessary

The grammar of graphics

- A useful way to think about statistical graphics is Wilkinson's grammar of graphics
- It provides a grammar that we can use to understand and construct graphics
- The grammar has several components which underpin any statistical graphics

Components of the grammar

- **Data** and a set of mappings to describe how variables should be mapped to perceivable aesthetic attributes
- **Geometric** objects which will appear in the plot e.g., points, lines, polygons
- **Statistical transformations** which may be used to change or summarise the data
- **Scales** which map values in the data space to values in the aesthetic space e.g., shape, colour, size
- **Coordinate system** to govern how coordinates in the data are mapped to the plane of the graphic
- **Faceting** which splits the data into subsets

Graphics in R

- R has a number of graphics systems
- Some are in Base R and some were introduced through packages such as lattice
- These days, *ggplot2* is the *de facto* standard for graphics in R and is one of the greatest strengths of R

- It is also part of the tidyverse so will it will fit nicely with our approach
- The *gg* part stands for grammar of graphics
- This approach allows us to construct almost any type of statistical graphic by adding layers
- I will be drawing on Hadley Wickham's *ggplot2: Elegant graphics for data analysis*

Just because you can do something doesn't mean you should

- As in language, we can say something which is grammatically correct but which doesn't make sense
- Consider the following sentence:

> The angry rock barked like a comma

- The sentence is grammatically correct but has no meaning
- The ggplot2 approach aims to let you construct whatever you want
- Always think about what you are trying to show and what makes sense
- The grammar doesn't help us fine-tune the appearance but this is something which is worth taking the time to do

ggplot2

- ggplot2 must be loaded, but it will load with the rest of the tidyverse if you use *library(tidyverse)*
- There are several options for how we can write a plot
- We'll pick a common approach (which is the way I do it too)
- The plot is constructed by adding elements
- We begin a plot with a call to the *ggplot()* function
- Note that the function is ggplot and the package is ggplot2
- We don't have to specify any arguments at this point, but I will suggest we do

Arguments in ggplot

- The first argument is the data which will be used
- This could be piped in or written in the function
- I prefer to pipe the data in

- Within the ggplot function we can specify aesthetic mappings with the *aes* function
- We start with an x variable (shown on the horizontal axis)
- We can also specify a y variable (on the vertical axis)
- Other mappings can be supplied here but we will leave that for now

Adding a geometry

- Having named the data and supplied the aesthetic mappings, we can specify the geometry to use
- This will depend on what we are trying to show
- We will explore several types today

Our dependent variable

- A good place to start is by looking at our dependent variable
- There are a limited number of plots which we can make with only one variable
- We will start by looking at two important examples: *histograms* and *density plots*
- Both of these will allow us to examine the *distribution* of the variable
- This is interesting in itself, but later we will see it is important for other reasons

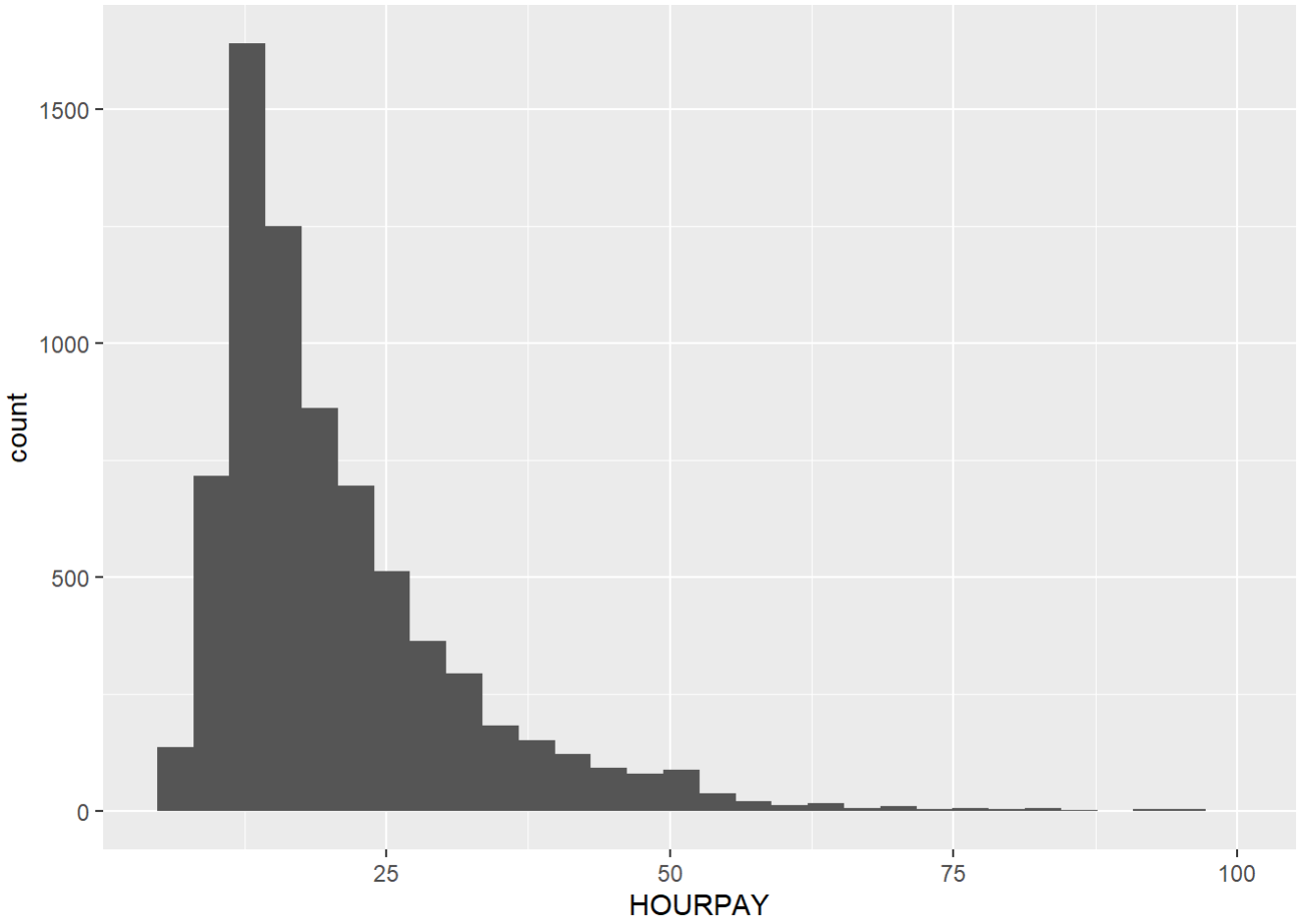
Histograms

- A histogram splits a continuous variable into a number of discrete bins
- The number of observations in each bin is counted
- Columns/bars are used to show the number of observations in each bin
- We can choose the size of the bins, although most software will supply a default
- We will be displaying the data with columns, however first we would have to aggregate the data in bins
- Because histograms are so important, ggplot has a shortest
- we can use this by using the histogram geometry
- This will take care of the aggregation and display for us

Distribution of hourly wage

```
lfs %>% ggplot(aes(HOURPAY)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



What can way say about the histogram

- Firstly, we can see that the distribution is *positively skewed*
- The bulk of the observations are at the lower end with some high earners on the far right
- There are a number of ways we can improve the appearance of the plot
- Importantly, most of the plot space is empty because of the distribution of the data
- Before dealing with this problem, we will learn a few ways to improve the plot

Labels

- Any plot you present **must** be properly labelled
- You should not leave raw variable names, particularly when the variable name is not very informative
- There are various ways of labelling a plot
- I suggest adding the *labs* function to a plot
- We can specify the element we want to label with this function

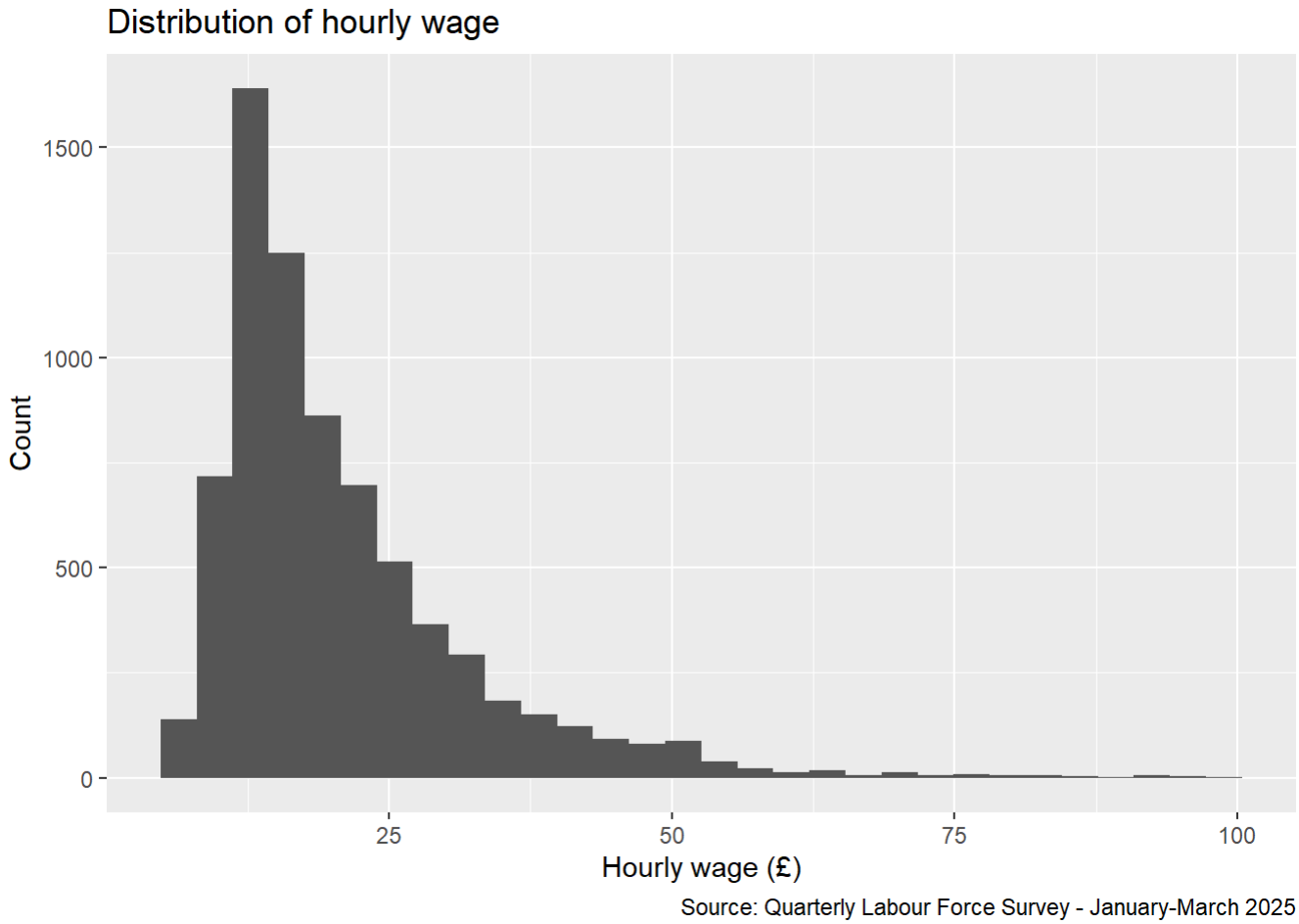
Some arguments of the labs function

- title: the title to place over the plot
- subtitle: the subtitle to appear under the title
- x: the label for the horizontal axis
- y: the label for the vertical axis
- caption: a caption to appear at the bottom right of the figure. This may contain notes such as the data used to make the plot

Adding labels to the histogram

```
lfs %>% ggplot(aes(HOURPAY)) +
  geom_histogram() +
  labs( title = "Distribution of hourly wage",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025")
```

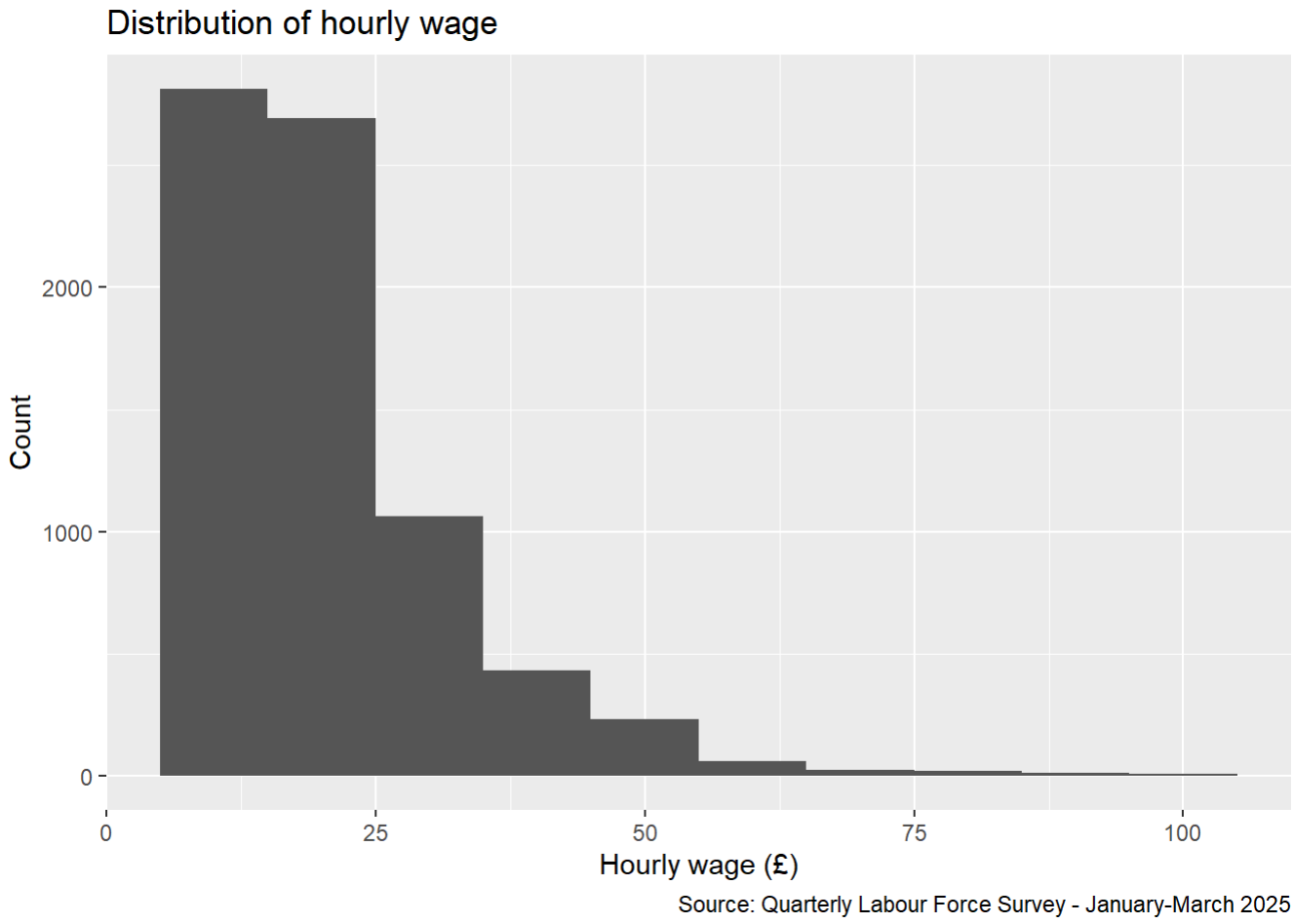
Labelled histogram



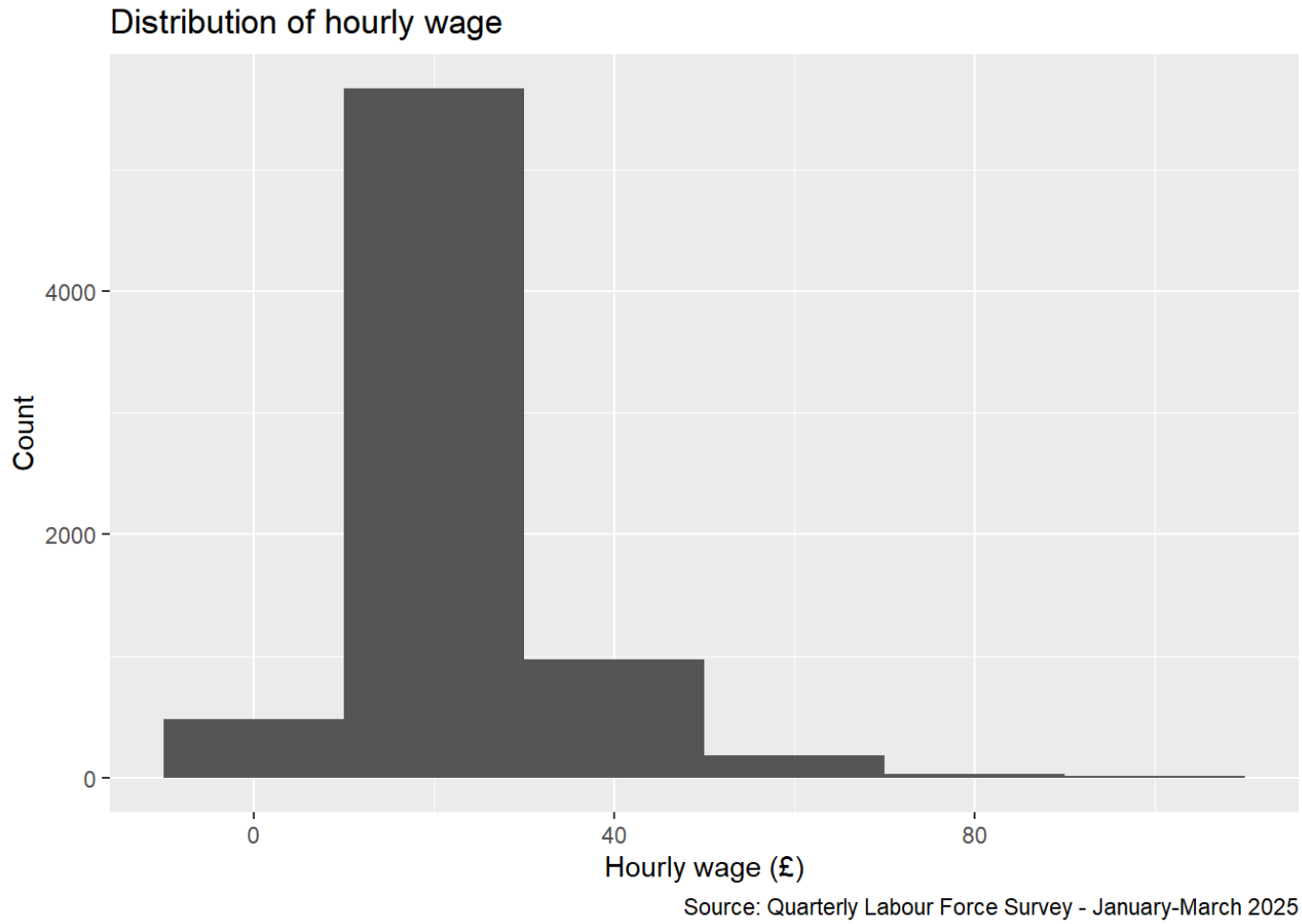
Bin width of £10

```
lfs %>% ggplot(aes(HOURPAY)) +
  geom_histogram(binwidth = 10) +
  labs( title = "Distribution of hourly wage",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025")
```

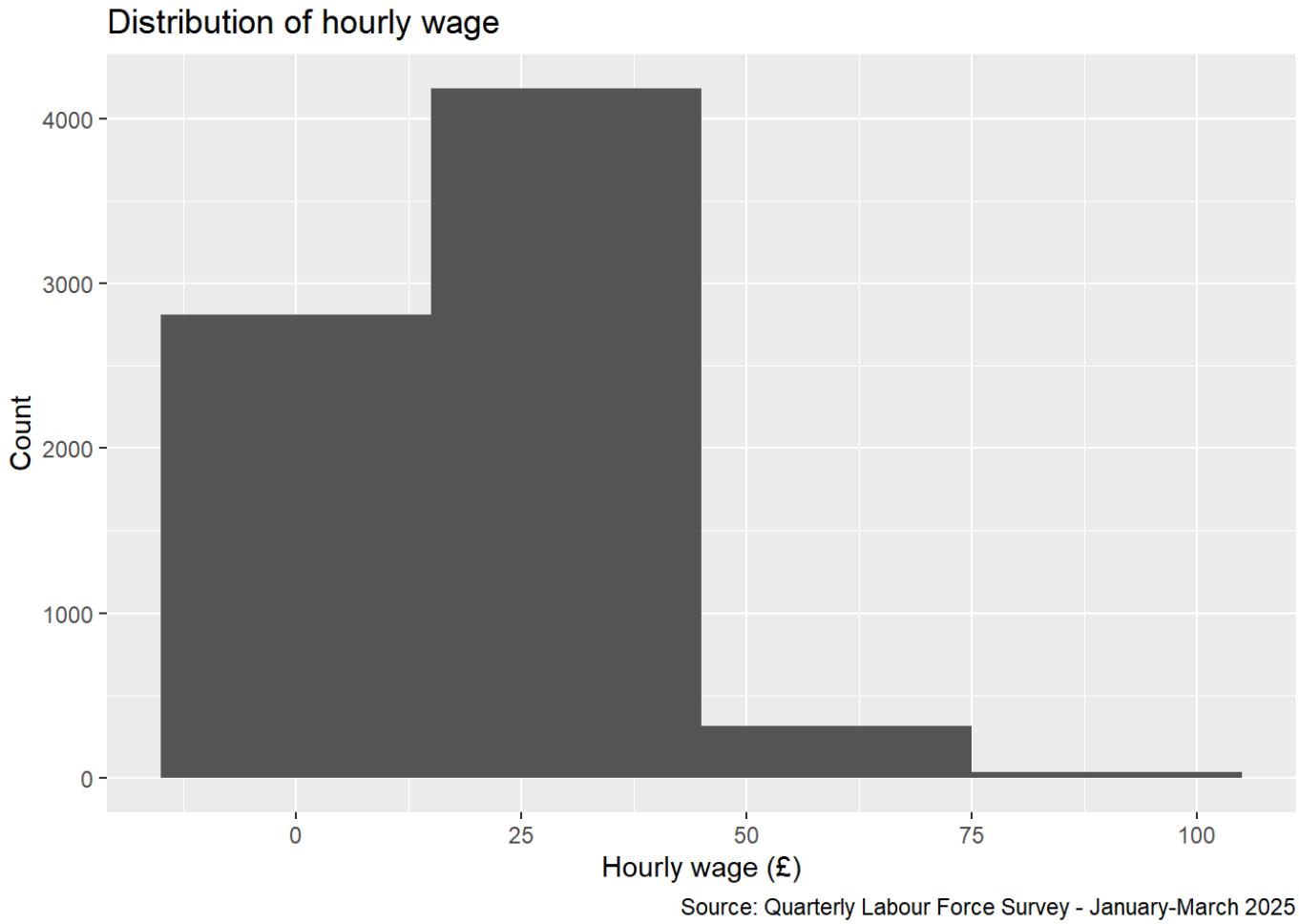
Bin width of £10



Bin width of £20



Bin width of £30

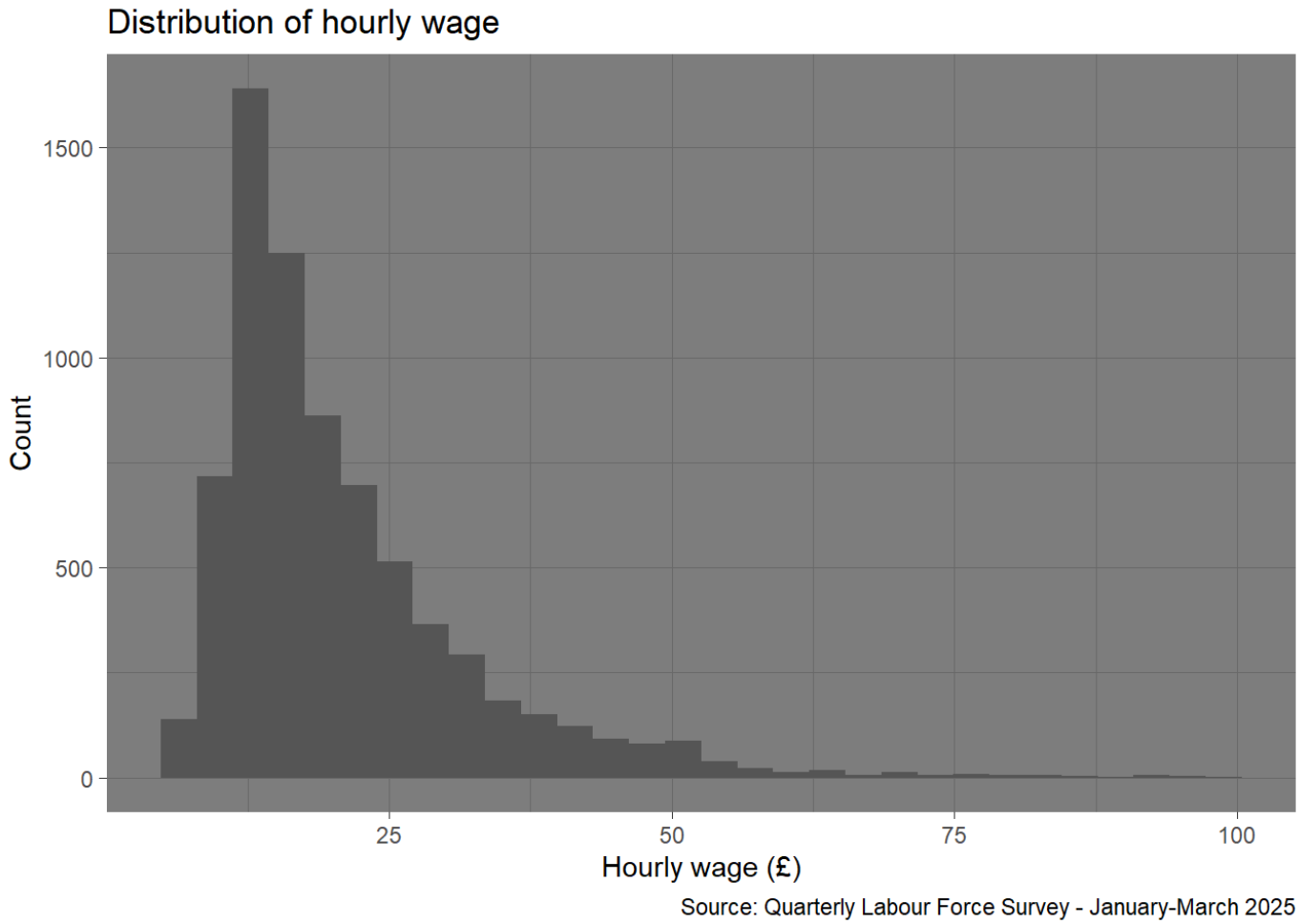


Other options

- You can customise pretty much anything about the plot's appearance
- You can also choose the theme of a plot, where someone else has already done some of the design work
- For example, you can choose *theme_dark()*, *theme_bw()*, or *theme_void()*
- You can look at the *ggthemes* package for more options
- There are other packages with all sorts of themes # Adding a theme

```
lfs %>% ggplot(aes(HOURPAY)) +
  geom_histogram() +
  labs( title = "Distribution of hourly wage",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_dark()
```

Histogram with the dark theme



Dealing with the skew

- Most of the plot space is taken up with empty space
- This is probably cause by a few high values
- If we consider these values to be erroneous, we might consider excluding them
- It is unwise to exclude observations unless we have good reason to
- We might consider excluding them from the plot
- We can do this either by filtering the data or by limiting the plot range

Checking for highly paid people

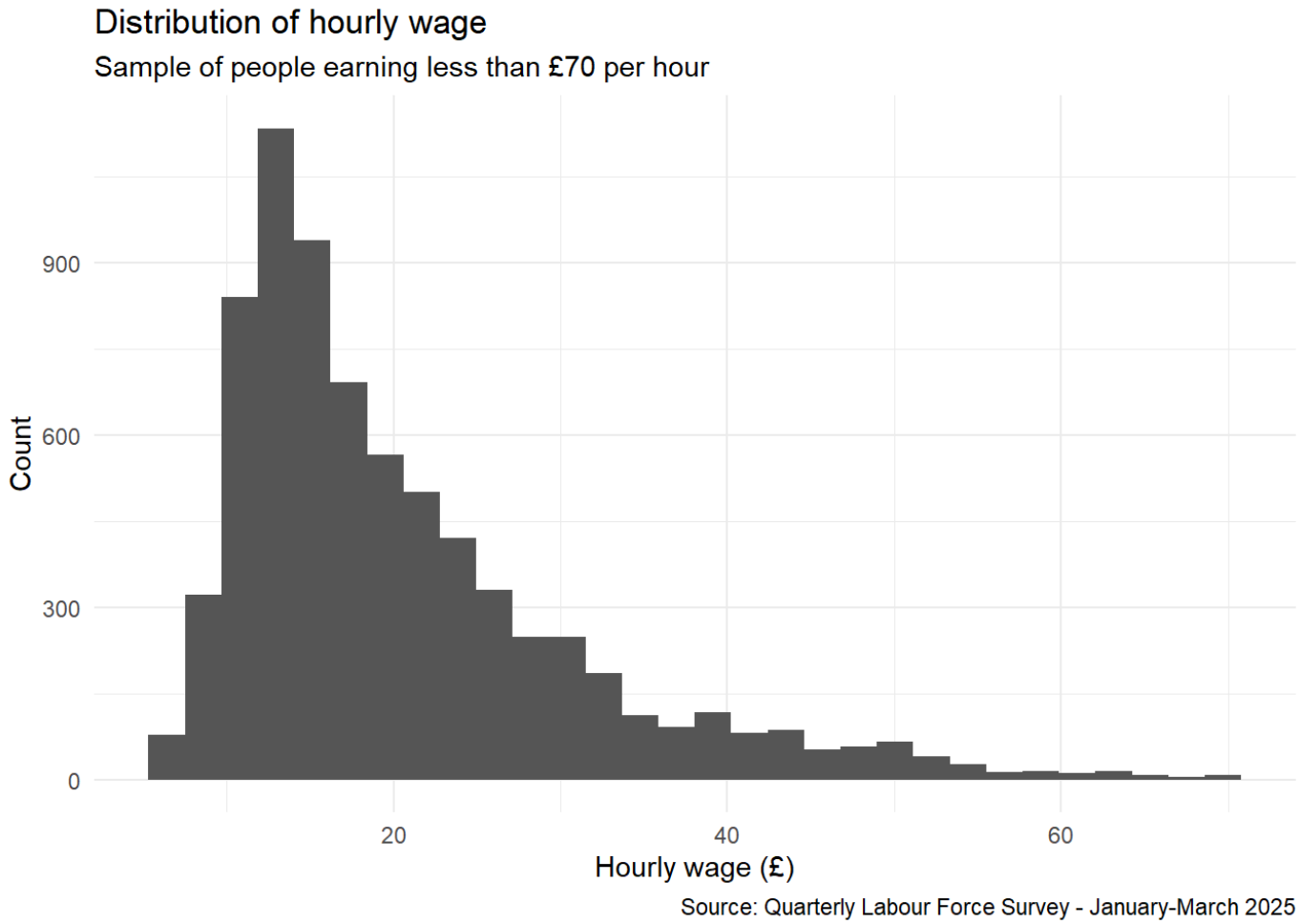
```
lfs %>% arrange(-HOURPAY) %>% select(HOURPAY) %>% head(n=10)
```

```
## # A tibble: 10 × 1
##    HOURPAY
##      <dbl>
##  1    98.8
##  2    96.2
##  3    96.2
##  4    96.1
##  5    95.4
##  6    94.0
##  7    93.8
##  8    93.6
##  9    92.3
## 10    91.4
```

Histogram for people earning <£70 per hour

```
lfs %>% filter(HOURPAY<70) %>%
   ggplot(aes(HOURPAY)) +
  geom_histogram() +
  labs( title = "Distribution of hourly wage",
        subtitle = "Sample of people earning less than £70 per hour",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal()
```

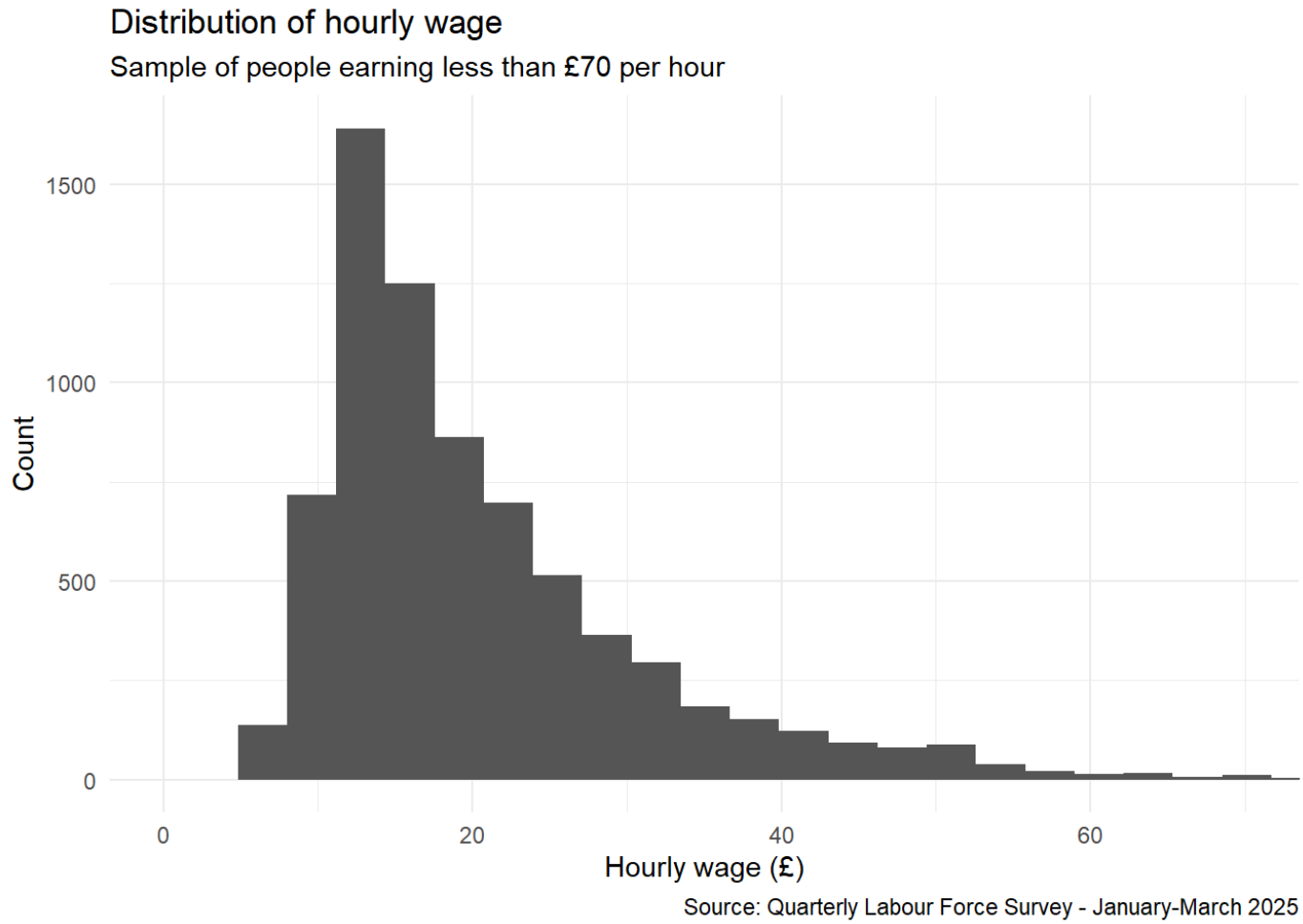
Histogram for people earning <£100 per hour



Alternative approach

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_histogram() +
  labs( title = "Distribution of hourly wage",
        subtitle = "Sample of people earning less than £70 per hour",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  coord_cartesian(xlim = c(0,70))
```

Alternative approach



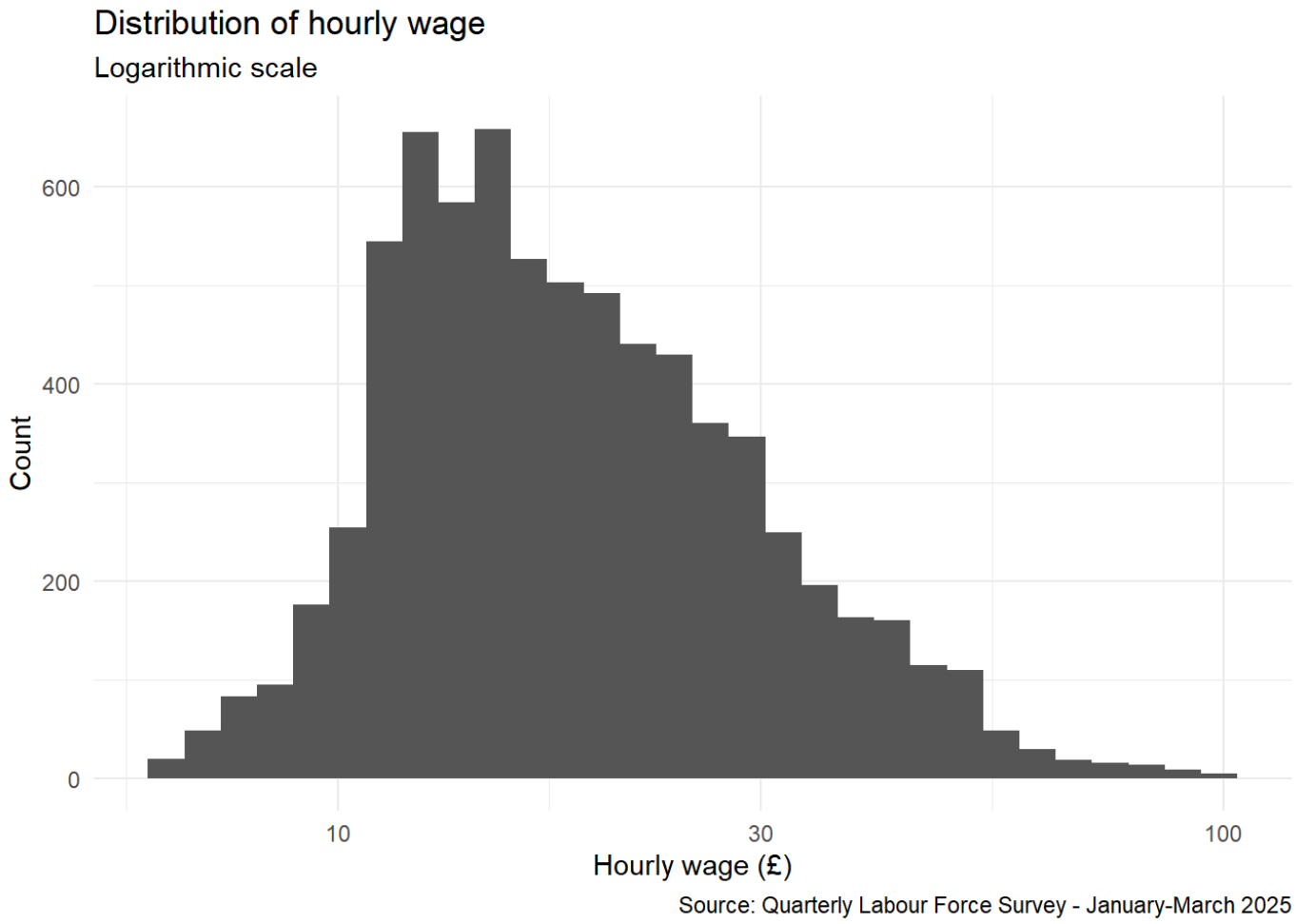
Linear scales versus logarithmic scales

- Our horizontal axis uses a linear scale
- As we move along the axis, the numbers increase at a constant rate
- This is not well suited to situations where the maximum values is substantially larger than the smallest value
- As we move along logarithmic scale, the numbers increase at an increasing rate
- The distance between 1 and 10 will be the same as the distance between 10 and 100 which will be the same as the distance between 100 and 1000
- This allows us to display a large range on a small plot
- Note that it won't work if we have zeros in our data because the log of 0 doesn't exist

Histogram with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_histogram() +
  labs( title = "Distribution of hourly wage",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10()
```

Histogram with a logarithmic scale



Density plots

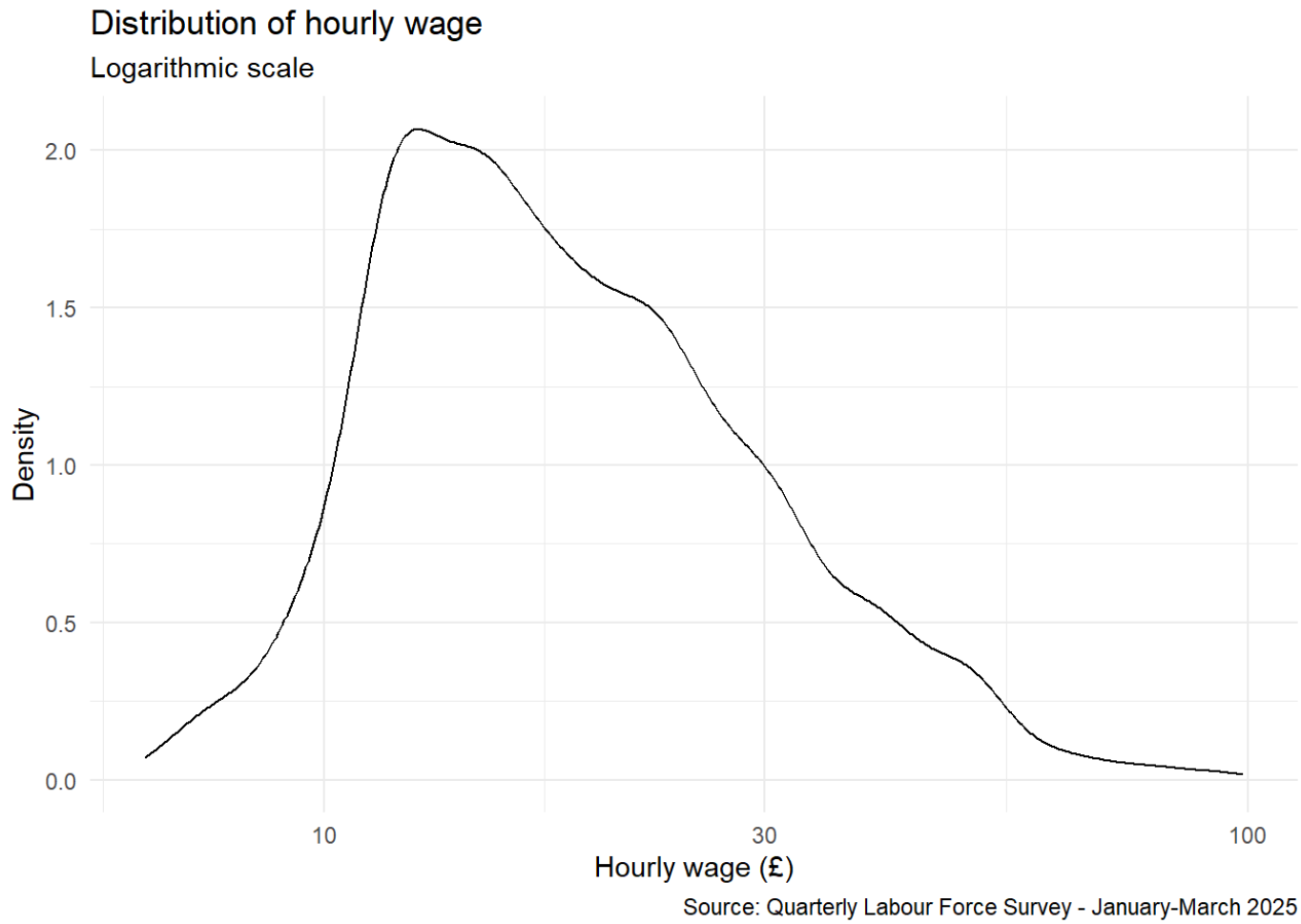
- The *density plot* is closely related to the histogram and tells us much the same thing
- Instead of splitting the data into bins, a kernel smoother is applied to smooth out noise
- This can help us see the underlying distribution
- It avoids some of the problems which can be caused by having an inappropriate number of bins

- It is produced in a very similar way to a histogram

Density plot with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_density() +
  labs( title = "Distribution of hourly wage",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10()
```

Density plot with a logarithmic scale



Adding a rug to the plot

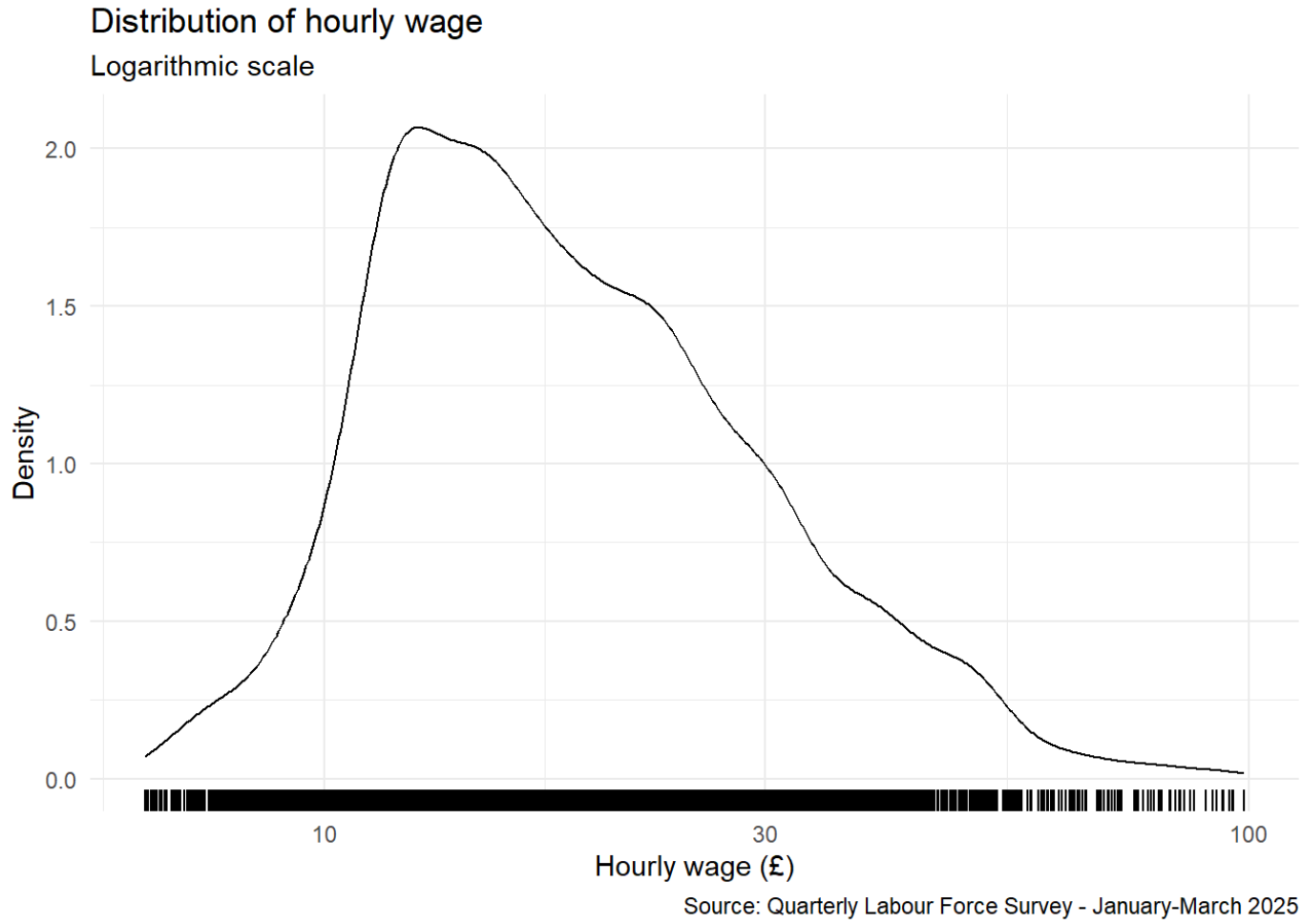
- Adding a rug to a plot adds a bar at the bottom of the plot
- The bar contains a vertical line indicating where each data point sits

- It doesn't take up much additional space but adds some extra information about the distribution
- We can add a rug by adding *geom_rug* to the plot

Density plot with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_density() +
  labs( title = "Distribution of hourly wage",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```

Density plot with a logarithmic scale



What should we be looking for?

- The histogram and the density plot have shown us much the same
- There is a clusterings of observations around the mean
- There is only one peak in our distribution
- Sometimes there might be more than one peak. Then we would want to know the reason
- We should also keep an eye out here for any kind of anomoly which doesn't make sense
- The distribution is highly skewed i.e., some people are earning very high hourly wages

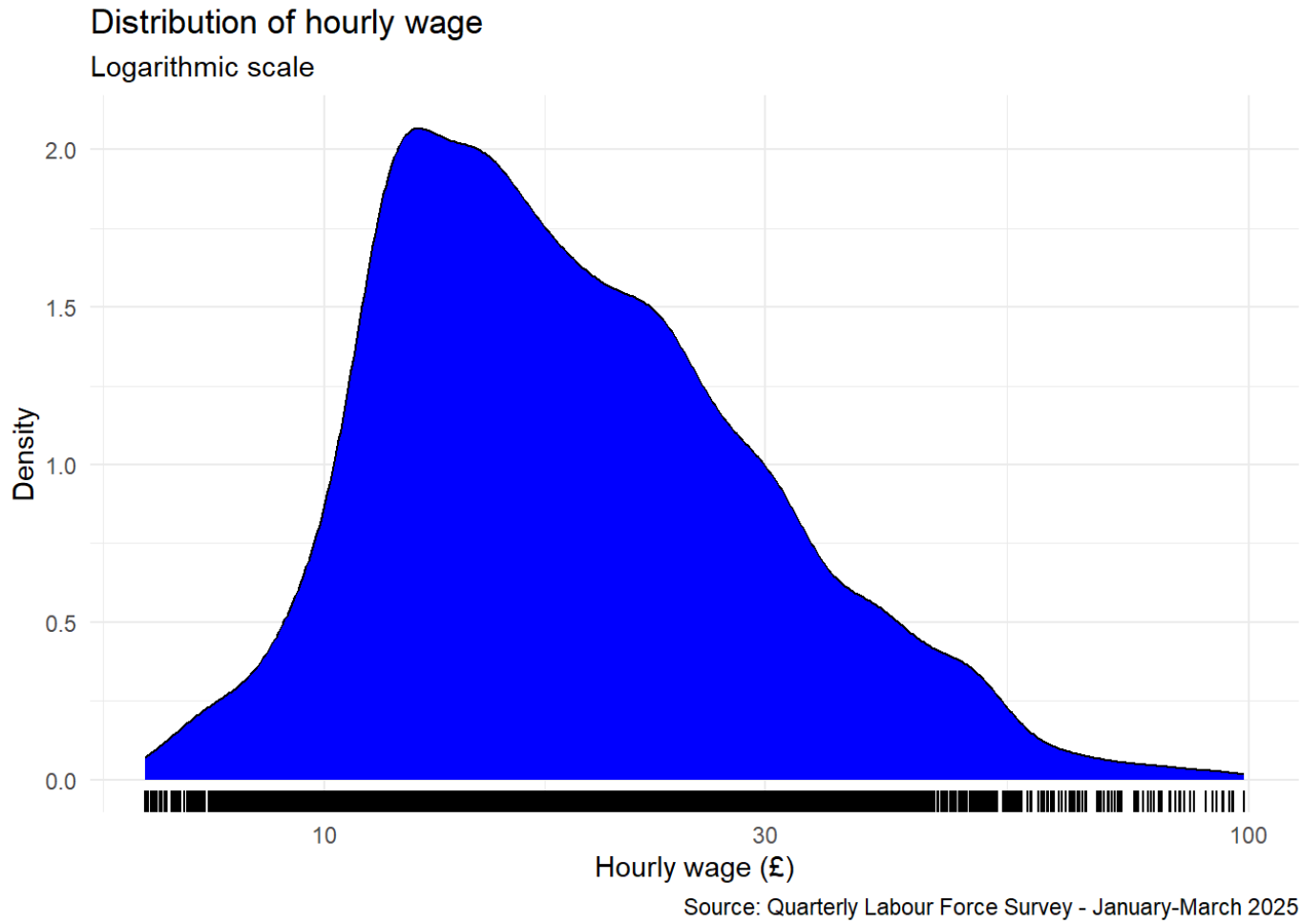
Adding to our density plot

- We can colour in the area under our density curve if we want
- Suppose we want to colour it in blue

Density plot with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_density(fill="blue") +
  labs( title = "Distribution of hourly wage",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```

Density plot with a logarithmic scale



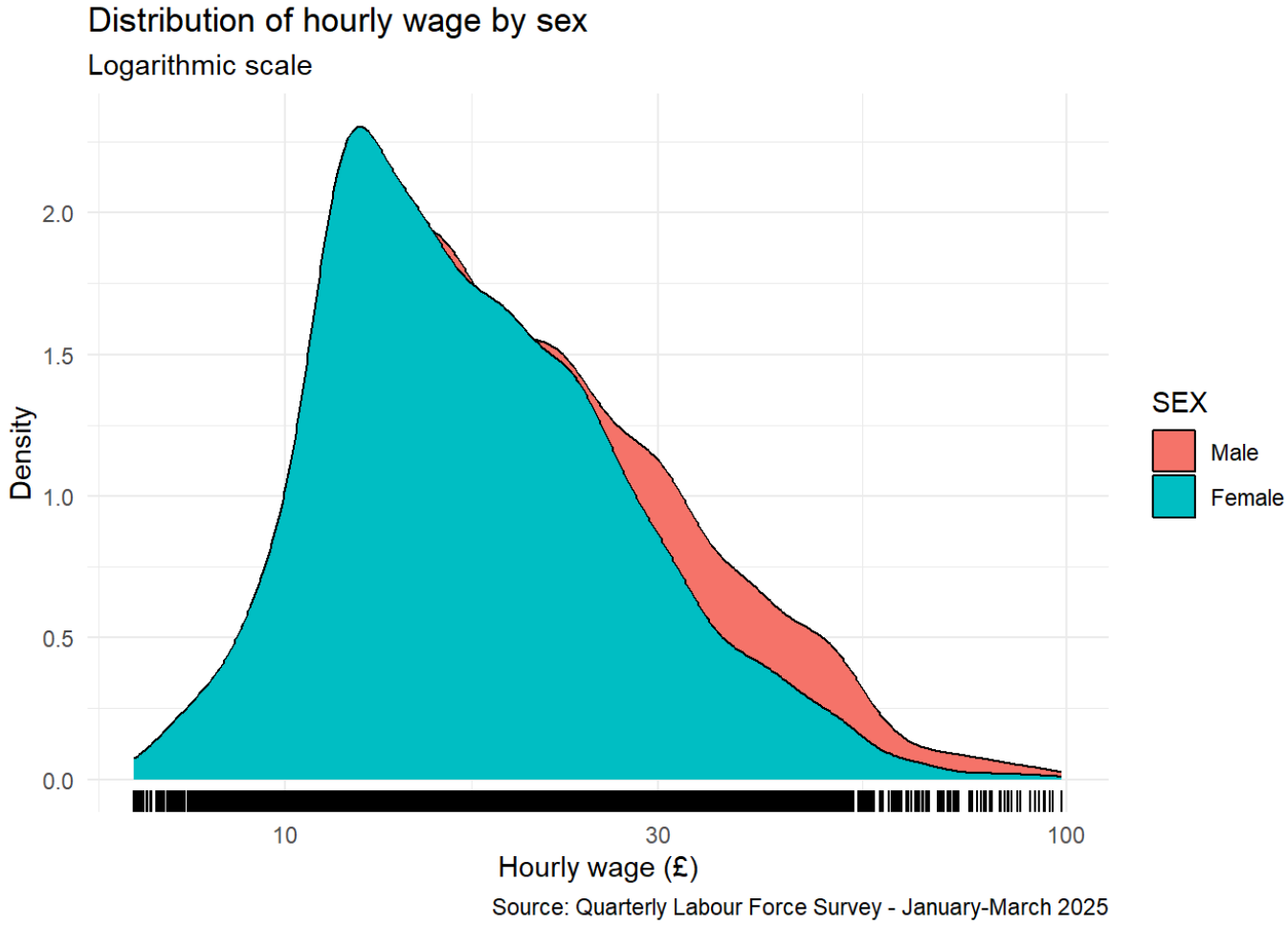
Adding another variable

- Colouring in the plot in is nice but it doesn't tell us anything else
- What we can do is have more than one density curve
- Then we can colour them in according to which group the observations belong to
- In our case, we can use the SEX variable to split
- We have to add this information to our aesthetic mapping
- We can do this either in the call to *ggplot* or to *geom_density*

Density plot with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_density(aes(fill=SEX)) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```

Density plot with a logarithmic scale



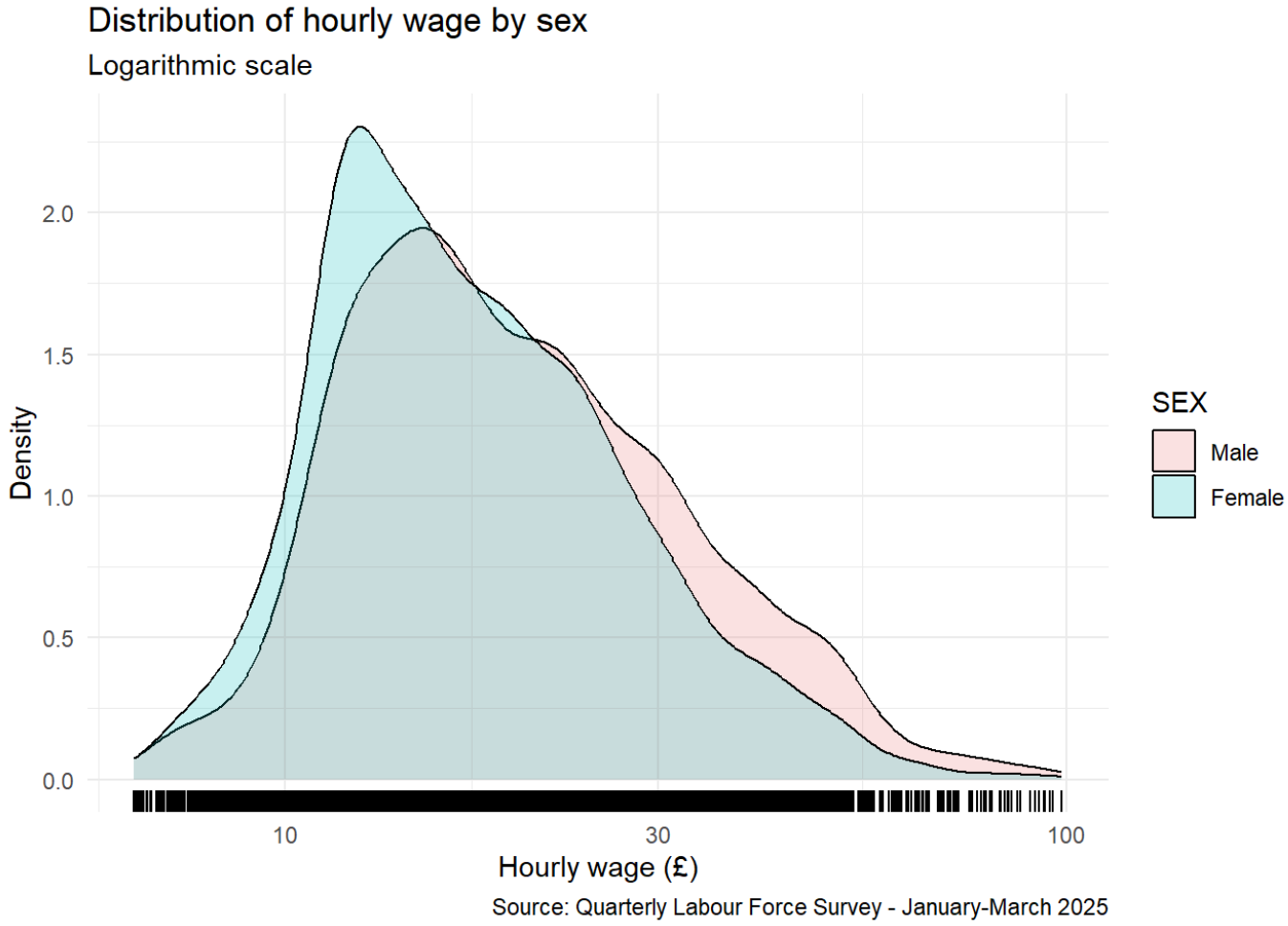
Improvements

- This is quite informative, but the curves are overlapping
- We can remedy this situation by reducing the opacity of the filled areas
- This can be achieved by setting the *alpha* parameter

Density plot with a logarithmic scale

```
lfs %>%
   ggplot(aes(HOURPAY)) +
  geom_density(aes(fill=SEX), alpha=0.2) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```

Density plot with a logarithmic scale



The legend

- The plot legend explains to the reader what is being shown
- We can customise this if we want
- We should limit the number of groups we have in a plot
- We can really only have about 8 clearly distinguishable colours

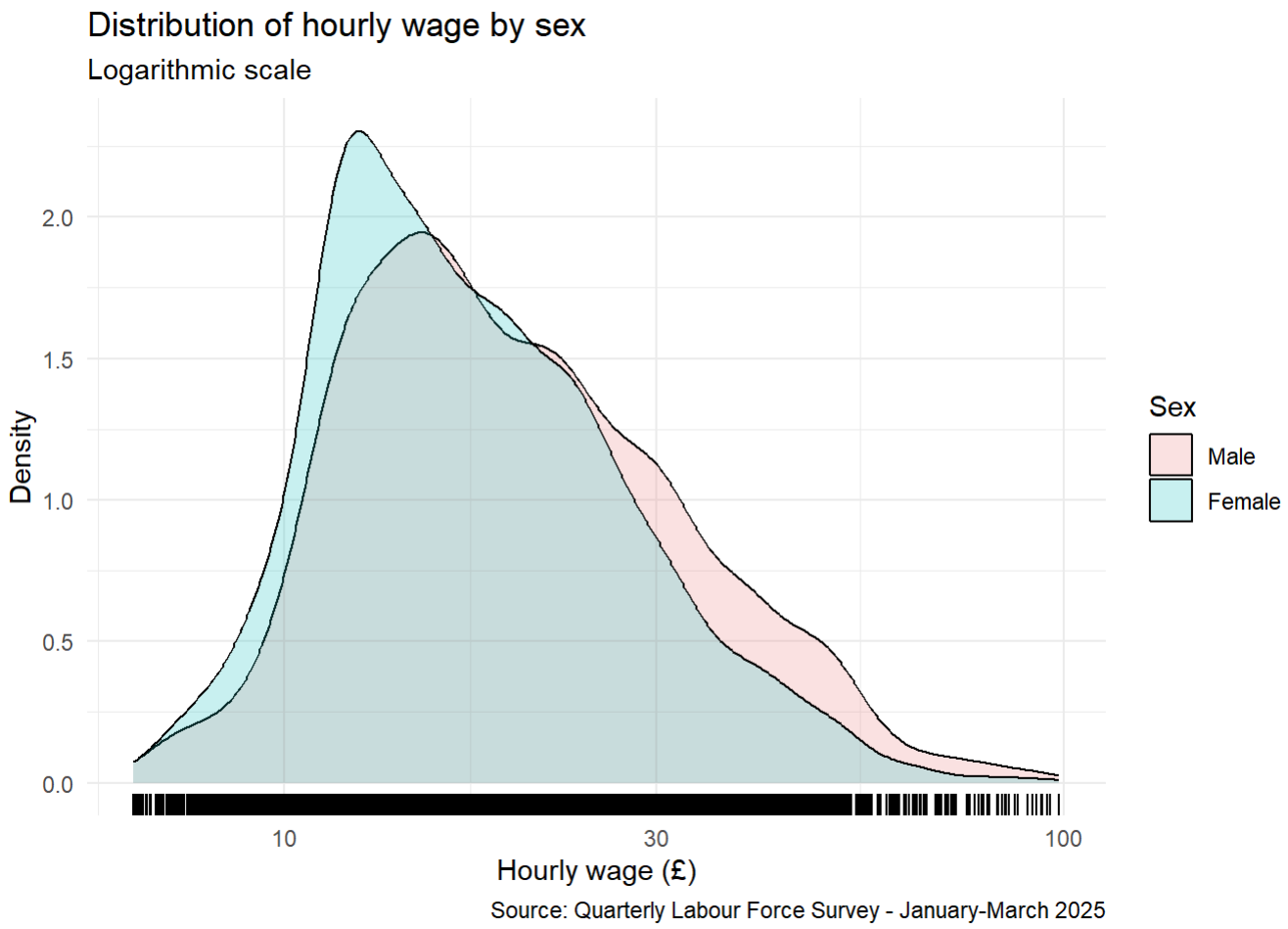
- Colour choice is important. ggplot2 tried to choose sensible colours by default
- There are several other packages with a variety of colour palates e.g., *viridis* or *ggthemes*
- Consider people with colour deficient vision (about 4.5% of the UK population - mostly male)

The legend

- Our legend has used the variable name SEX to label the legend
- This is at least a meaningful name, however it shouldn't be capitalised
- In this case, from the title of the plot and the labels of the categories, it is fairly obvious what the legend shows
- We could therefore consider dropping the legend title
- The legend can be customised using the labs() function
- We can also use the theme function

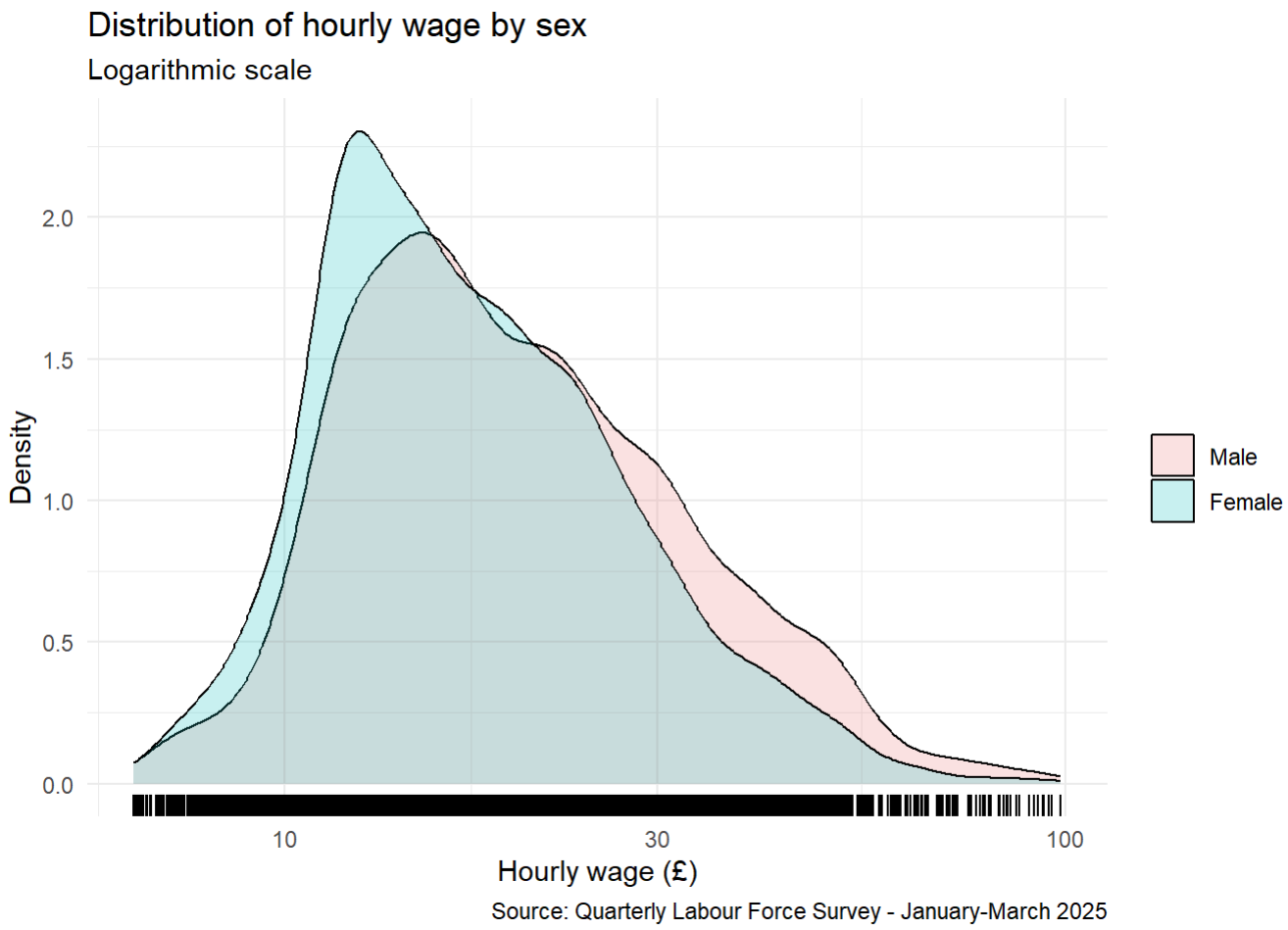
Changing the legend label

```
lfs %>%
  ggplot(aes(HOURPAY)) +
  geom_density(aes(fill = SEX), alpha = 0.2) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        fill = "Sex",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```



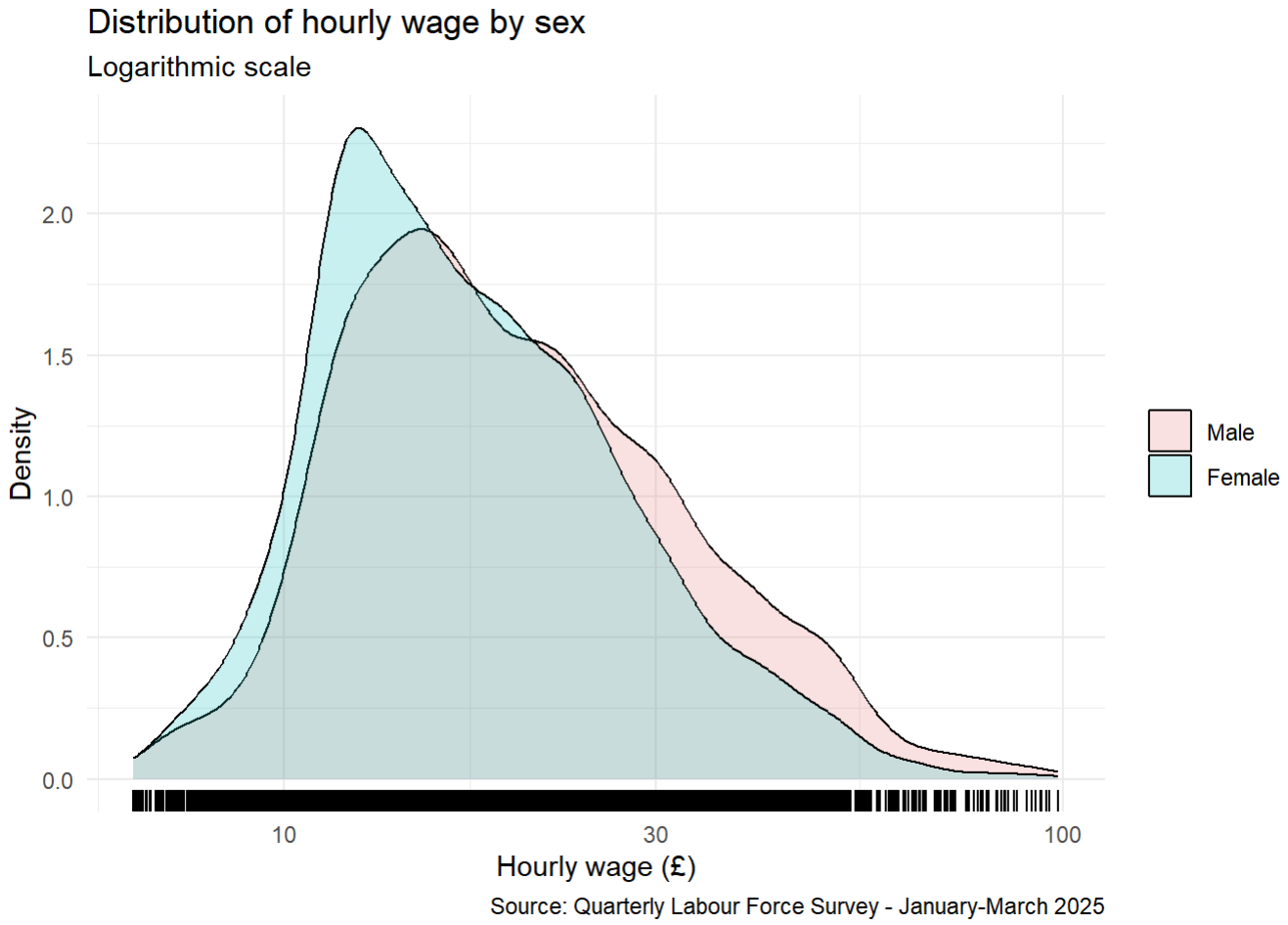
Hide the legend label (sort of cheating)

```
lfs %>%
  ggplot(aes(HOURPAY)) +
  geom_density(aes(fill = SEX), alpha = 0.2) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        fill = "",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug()
```

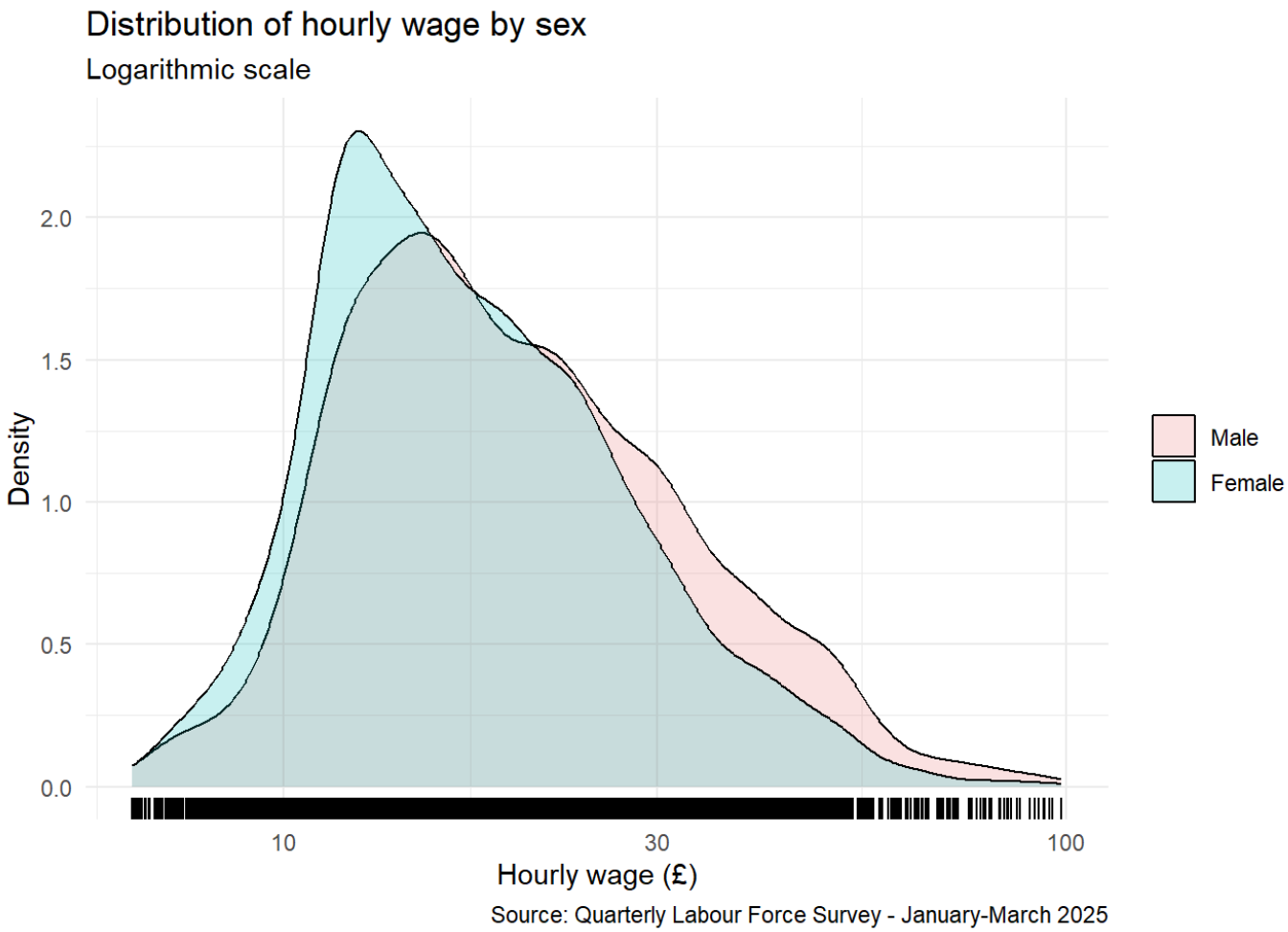


Hide the legend label (proper way)

```
lfs %>%
  ggplot(aes(HOURPAY)) +
  geom_density(aes(fill = SEX), alpha = 0.2) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Density",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  geom_rug() +
  theme(legend.title = element_blank())
```



Density plot with a logarithmic scale



Changing the fill on the histogram

- The fill attribute can be used for many different types of plots
- Let's try it out on our histogram

Histogram with a logarithmic scale

```
lfs %>%
  ggplot(aes(HOURPAY)) +
  geom_histogram(aes(fill=SEX)) +
  labs( title = "Distribution of hourly wage by sex",
        subtitle = "Logarithmic scale",
        x = "Hourly wage (£)",
        y = "Count",
        caption = "Source: Quarterly Labour Force Survey - January-March 2025") +
  theme_minimal() +
  scale_x_log10() +
  theme(legend.title = element_blank())
```

Histogram with a logarithmic scale



Summary

- Today we have meet one of R's most powerful graphics systems
- We have also covered the key building blocks of making a compelling data visualisation
- Next time we will develop what we covered today and see what other wonders we can create